

# Turbofan

Aerospace Propulsion

Laboratory Session 3



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# 1 Introduction

Modern civil turbofan engines are designed to operate efficiently at a specific cruise condition, yet in real operation they must perform reliably over a wide range of flight regimes, including take-off, climb, descent, and off-design ambient conditions. Understanding how engine performance shifts away from the design point is critical for aircraft performance, fuel efficiency, safety margins, and lifecycle considerations.

This laboratory session focuses on the thermodynamic and aerodynamic behaviour of a high-bypass turbofan engine when transitioning from its design cruise condition to off-design operating points, with special emphasis on take-off performance. Using the PROOSIS simulation environment, a reference engine was first sized at its cruise condition by imposing thrust requirements and allowing geometric parameters, such as nozzle exit areas, to converge. This established a consistent baseline for subsequent analyses.

The study then explored the sensitivity of engine performance to key design and operational parameters, including fan pressure ratio ( $\pi_f$ ), bypass ratio (BPR), nozzle velocity matching, and ambient conditions. Parametric sweeps were conducted to evaluate the trade-offs between specific thrust and specific fuel consumption, highlighting the fundamental compromises that govern turbofan design.

Finally, off-design simulations were carried out to quantify how real engines respond to changes in flight Mach number, altitude, and ambient temperature. These results provide insight into the physical limitations imposed by mass flow variation, thermal constraints, and component maps, and demonstrate how performance deteriorates or improves depending on operating environment. The overall objective is to link theoretical turbomachinery concepts with realistic engine behaviour under operationally relevant conditions.

## 2 Turbofan Design for a Civil Aircraft

### 2.1 Sizing of Engine Nozzles

In this part we are asked to build the design point of the engine at maximum cruise conditions. Using the thrust,  $F_N$ , as a boundary condition, we let PROOSIS compute the required nozzle areas: **A9** and **A19**.

After having computed this, we were asked to add another condition: velocity matching at the nozzles.

$$v_9 = v_{19}$$

Once the core and bypass jet velocities matched, we observed the changes in the areas, net thrust and other variables shown in the table below.

| Experiment  | A19 ( $m^2$ ) | A9 ( $m^2$ ) | V19 ( $m/s$ ) | V9 ( $m/s$ ) | S ( $g/kN \cdot s$ ) | sFn ( $m/s$ ) |
|-------------|---------------|--------------|---------------|--------------|----------------------|---------------|
| No Matching | 3.61482509    | 1.09570385   | 317.880268    | 493.653738   | 13.5951021           | 187.396251    |
| Matching    | 3.62468714    | 1.6006656    | 319.918387    | 319.918387   | 13.1164134           | 180.160263    |

## 2.2 Variation of $\pi_f$ and $\alpha$ (BPR)

The objective of this exercise was to observe the reaction of the engine when a double parametric experiment was performed between:

$$\text{Fan pressure ratio: } \pi_f \in [1.1 \quad 2.5]$$

$$\text{Bypass ratio: } BPR \in [2 \quad 12]$$

The idea is to see how these two variables affect fuel consumption (S), specific thrust (sFn), and other performance quantities.

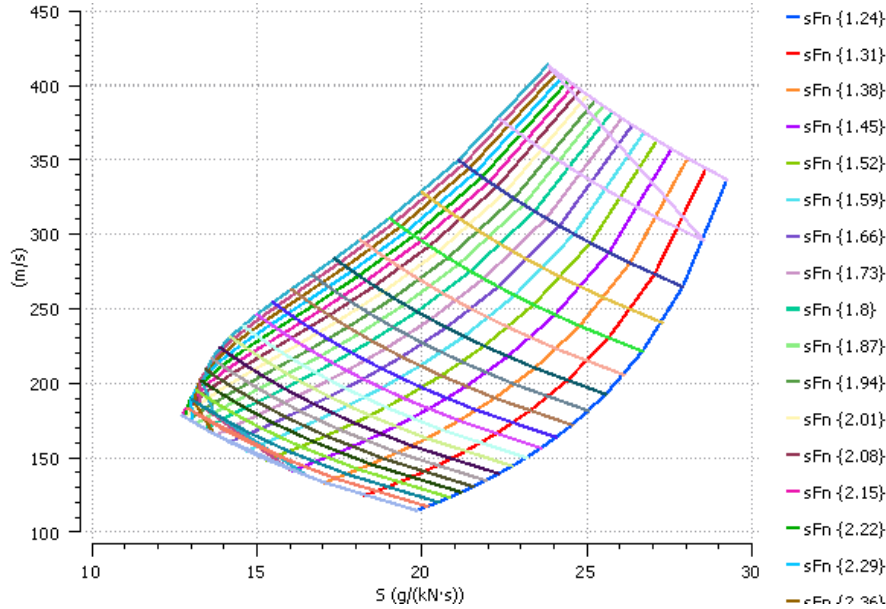


Figure 1: Carpet plot S vs sFn

The figure above shows the double parametric study between  $\pi_f$  and  $BPR$ . As both variables are directly related to the specific thrust and specific fuel consumption, we can see that while increasing  $\pi_f$  raises S and sFn, increasing  $BPR$  reduces them. The plot highlights the design trade-off between efficiency and thrust production.

## 2.3 Effort of $\pi_f$

In this exercise, the already sized engine in Exercise 1 is being reused. We performed another parametric study, this time focusing only on the fan pressure ratio under cruise conditions, and fixed the nozzle geometry to the one obtained in the first part without the velocity matching. We then analysed some quantities with respect to  $\pi_f$  to see the performance of the engine. This way we are able to see how sensitive the engine is to this parameter when operating at cruise.

| <b>A19 (<math>m^2</math>)</b> | <b>A9 (<math>m^2</math>)</b> | <b>V19 (<math>m/s</math>)</b> | <b>V9 (<math>m/s</math>)</b> |
|-------------------------------|------------------------------|-------------------------------|------------------------------|
| 3.61482509                    | 1.09570385                   | 317.889268                    | 468.757381                   |

Table 1: Areas and Velocities Comparison

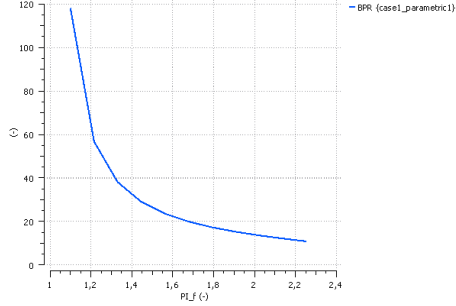


Figure 2: Bypass Ratio

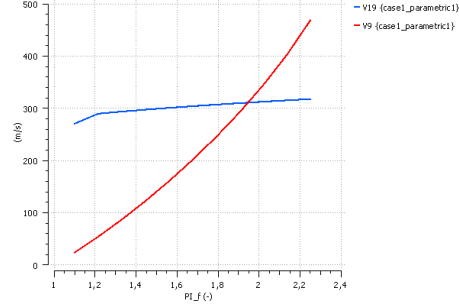


Figure 3: Velocity Variation

The first change we notice is in the bypass ratio. As  $\pi_f$  increases, the BPR drops sharply. A higher fan pressure ratio means the fan is doing more work, and this pushes more mass flow toward the core, reducing the proportion that goes through the bypass stream.

This relates to the velocity plot, which shows that  $V_9$  (core jet velocity) grows significantly with  $\pi_f$ , while  $V_{19}$  (bypass velocity) stays almost flat. The bypass stream barely reacts unless the nozzle geometry changes, whereas the core flow is directly affected by a stronger fan.

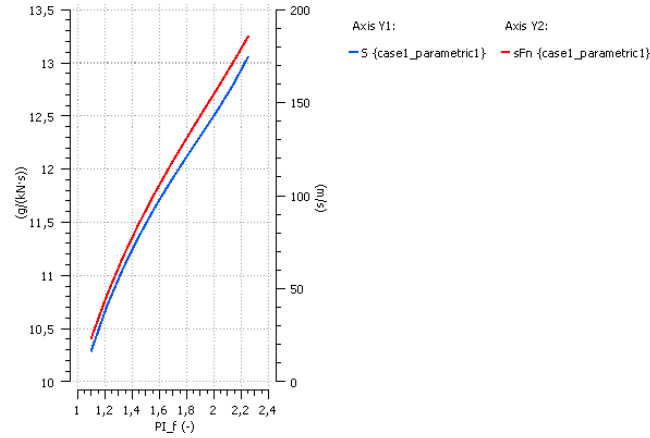


Figure 4: S and sFn

Fuel consumption  $S$  and specific thrust  $sFn$  also rise as  $\pi_f$  increases, although in different scales. This is the first sign that while  $\pi_f$  is increased so is the thrust, but the fuel consumption is also higher.

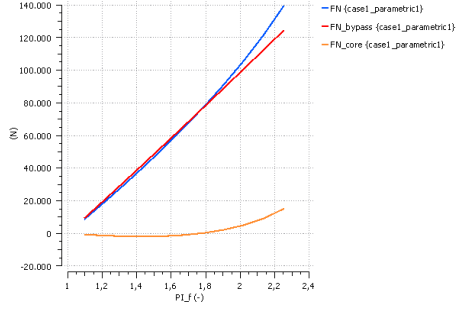


Figure 5: Net Thrust

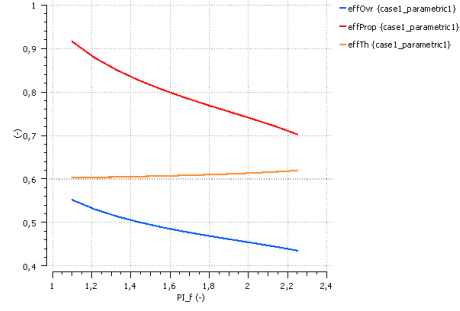


Figure 6: Efficiencies

Regarding thrust, both the total thrust and the bypass thrust increase almost linearly with  $\pi_f$ . The core thrust starts slightly negative at low  $\pi_f$  (typical in high-bypass engines at cruise) but becomes positive at higher values. This happens because the stronger the fan, the more mass flow through the core.

The efficiency plots also behave as expected. The overall and propulsive efficiencies decrease with increasing  $\pi_f$ . The drop in propulsive efficiency is the largest, since a higher fan pressure ratio produces a larger mismatch between jet and flight velocities while consuming larger amounts of fuel. Thermal efficiency barely changes, which makes sense because we did not modify the core pressure ratios or turbine temperatures and are still working in design parameters.

### 3 Turbofan Off-Design for a Civil Aircraft

#### 3.1 Take-off Steady Calculations

For this part we first recalculated the on-design sizing of the engine but with the MAP files and then proceeded to make off-design calculations, always restoring the values of the latter. This allows us to obtain a clear comparison of both conditions for the Turbofan's functionality.

| <b>A19</b> ( $m^2$ ) | <b>A9</b> ( $m^2$ ) | <b>V19</b> ( $m/s$ ) | <b>V9</b> ( $m/s$ ) | <b>S</b> ( $g/kN * s$ ) | <b>sFn</b> ( $m/s$ ) | <b>FN</b> ( $N$ ) | <b>Tt4</b> ( $K$ ) |
|----------------------|---------------------|----------------------|---------------------|-------------------------|----------------------|-------------------|--------------------|
| 3.69447814           | 1.1215506           | 317.880268           | 445.70587           | 12.8878069              | 183.355982           | 140000            | 1750               |

Table 2: On-Design Conditions (Cruise)

| <b>A19</b> ( $m^2$ ) | <b>A9</b> ( $m^2$ ) | <b>V19</b> ( $m/s$ ) | <b>V9</b> ( $m/s$ ) | <b>S</b> ( $g/kN * s$ ) | <b>sFn</b> ( $m/s$ ) | <b>FN</b> ( $N$ ) | <b>Tt4</b> ( $K$ ) |
|----------------------|---------------------|----------------------|---------------------|-------------------------|----------------------|-------------------|--------------------|
| 3.69447814           | 1.1215506           | 345.566312           | 303.405584          | 9.46309934              | 275.71306            | 490000            | 1961.73925         |

Table 3: Off-Design Conditions (Take-off)

From the two tables we can see how different the engine behaves at take-off compared to the cruise design point. As we want the engines design points to keep constant, we fix the exit areas  $A9$  and  $A19$  to the values obtained in the design experiment; although this doesn't prevent the jet velocities from changing. The bypass velocity  $V19$  increases only slightly while the core velocity  $V9$  decreases. The latter is a consequence of the quantity of mass flow the engine tries to push at take-off, which is much larger than at cruise conditions.

The increase in the specific thrust  $sFn$  is a consequence of the turbine and compressor maps acting at their limit during take-off. This is also why the temperature at the exit of the burner  $Tt4$  is higher too.  $Tt4$  being larger during take-off also implies a larger amount of energy per kilogram of air in this period, which allows the turbine and fan to deliver much more work.

Finally, we see a decrease in the value of the specific fuel consumption  $S$ . Even though the engine is burning more fuel during take-off, it also produces more thrust because of the larger  $Tt4$  and mass flow rate.

### 3.2 Ambient Pressure Variations During Take-Off

In this exercise we set the value of  $Tt4$  to the one obtained in the previous calculation and then vary the ambient temperature to see the engines' reaction.

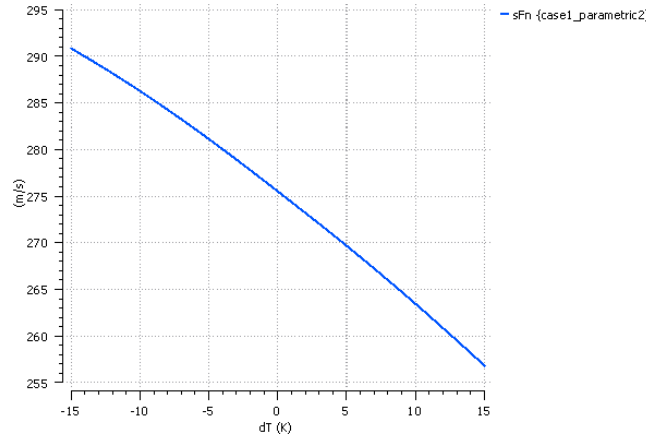


Figure 7: Specific Thrust

The behaviour of the specific thrust  $sFn$  with respect to ambient temperature  $dT$  is clear: as  $dT$  increases, we see an almost linear decrease in  $sFn$ . This happens because at higher ambient temperatures, the mass flow rate through the fan and core becomes smaller, which limits  $sFn$ .

### 3.3 Energy Perspective During Take-Off

Once we have a clear view of the engine's performance during take-off, we can also interpret the results from an energy point of view. The reduction in thrust with increasing ambient temperature can be explained by how the engine processes the incoming air. As the ambient temperature increases, there is a reduction in air density; therefore, the mass flow rate through the fan and core decreases.

Since the mass flow rate becomes smaller at higher temperatures, the engine receives less total energy per second from the combustion process. This translates into the reduction of kinetic energy, which implies a decrease in the exhaust jet velocities.

In simple terms, in hotter days the engine has less dense air to work with and cannot convert as much energy into useful thrust, resulting in poorer take-off performance.

### 3.4 Comparison of take-off, design, service ceiling and 0.1 mach results

For this analysis, two additional partitions were implemented in PROOSIS to evaluate the off-design behavior of the turbofan. In the Design and Take-off partitions, the prescribed maximum thrust was imposed as a boundary condition, while the flight condition variables  $h$ ,  $\Delta T$ , and  $M_0$  were released and treated as unknowns to allow the solver to adjust the operating point accordingly. A steady calculation was then performed for each of the four operating conditions to obtain the converged solutions shown in the table.

The results reveal clear performance differences across Design, Take-off, Service ceiling, and low-speed sea-level operation at  $M = 0.1$ . As expected for a high-bypass turbofan, the thrust levels depend strongly on ambient conditions, with Take-off producing the highest thrust (490 kN) (bounded value) due to high air density and large corrected mass flow, while the Service ceiling case produces the lowest thrust (99.9 kN) as a result of reduced density at altitude. The TSFC indicator  $S$  decreases substantially in the Take-off case relative to Design and Service ceiling conditions, reflecting improved fuel efficiency at higher thrust: the bypass stream generates a larger momentum change for only a moderate increase in fuel flow. Conversely, at Service ceiling, the drop in mass flow reduces thrust more significantly than it reduces fuel consumption, leading to a higher TSFC and reduced efficiency. The specific thrust  $sFn$  behaves consistently with physical expectations, reaching its highest value at Take-off and its lowest at Service ceiling, in line with variations in jet velocity and mass flow driven by atmospheric conditions. The  $M = 0.1$  sea-level case lies between Design and Take-off, as the high density increases available mass flow but the low flight speed limits ram compression. The slightly elevated turbine inlet temperature in the Take-off condition ( $Tt4 \approx 1766$  K) further increases turbine power and contributes to the exceptionally high bypass thrust observed.

All in all, the results obtained were sensible with high-bypass turbofan behavior: Take-off provides the highest thrust and best efficiency, Service ceiling suffers from low-density penalties, and the Design and low-speed sea-level points fall naturally between these extremes.

| Case            | FN (N)     | S (g/(kN·s)) | Tt4 (K)  | $\dot{m}_f$ (kg/s) | sFn (m/s) |
|-----------------|------------|--------------|----------|--------------------|-----------|
| Design          | 140000     | 12.888       | 1750.000 | 1.804              | 183.356   |
| Take-off        | 490000     | 7.850        | 1766.087 | 3.847              | 263.972   |
| Service Ceiling | 99901.928  | 12.888       | 1750.000 | 1.288              | 183.356   |
| M=0.1 SL        | 354571.294 | 7.743        | 1750.000 | 2.745              | 252.477   |

## 4 Conclusion

This laboratory successfully demonstrated how a civil high-bypass turbofan engine behaves when transitioning from its cruise design point to a range of off-design operating conditions. By first constructing a consistent design-point model, it was possible to isolate and understand the influence of both geometric and thermodynamic constraints on engine performance.

The nozzle sizing and velocity-matching exercises revealed the strong coupling between exit area, jet velocity, and overall efficiency. Enforcing equal core and bypass jet velocities reduced specific thrust but improved propulsive efficiency, illustrating the classic compromise between raw thrust capability and fuel economy. The parametric studies on fan pressure ratio and bypass ratio further highlighted this trade-off: increasing  $\pi_f$  increases thrust potential but degrades propulsive efficiency, while increasing BPR improves fuel efficiency at the expense of specific thrust.

Off-design analyses showed that take-off conditions significantly alter engine behaviour due to higher mass flow rates and elevated turbine inlet temperatures. The engine operates closer to component limits, producing substantially higher thrust while achieving lower specific fuel consumption owing to more effective momentum exchange in the bypass stream. Conversely, high-altitude and low-density conditions, such as the service ceiling case, penalise thrust capability and efficiency by reducing available mass flow and total energy addition.

The ambient temperature sensitivity study confirmed the physical dependence of thrust on air density, demonstrating that higher temperatures reduce mass flow and therefore limit thrust production. This directly explains real-world hot-and-high performance limitations experienced by aircraft.

Overall, the results were coherent with expected turbofan physics and provided a realistic picture of how design choices and environmental conditions govern performance. The exercise reinforced the importance of matching engine architecture to mission requirements, and showed how off-design analysis is essential for predicting real operational capability rather than relying solely on idealised design-point performance.

*Generative AI tools were used to assist in drafting and refining content, ensuring clarity and coherence. All outputs were reviewed and edited for accuracy and originality by the authors*